

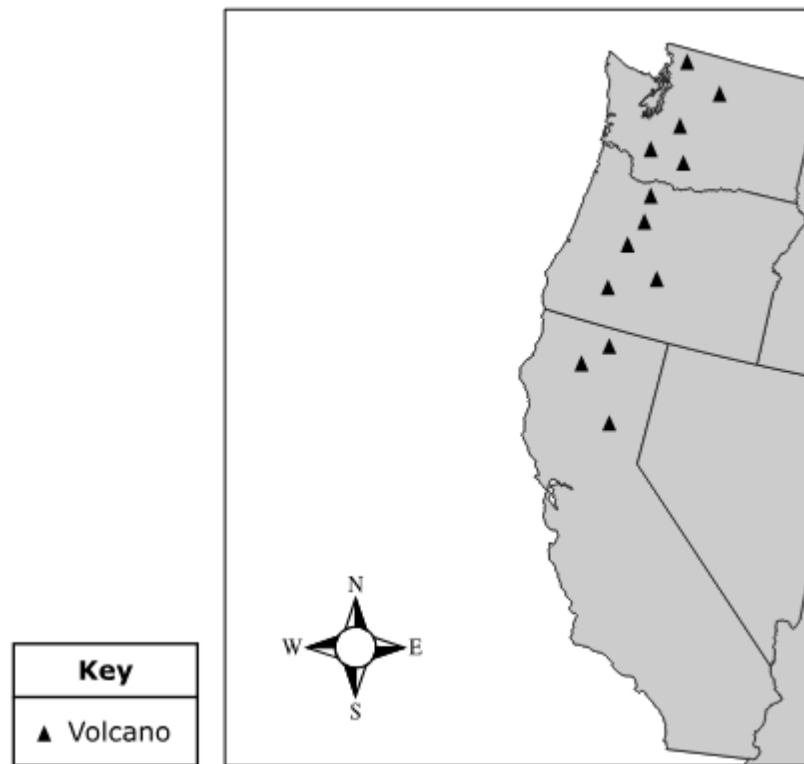
Tennessee Comprehensive Assessment Program

TCAP

Science
Grade 4
Practice Items



The map shows the location of some volcanoes in the United States.



What pattern is shown by the location of these volcanoes?

- M. Volcanoes are usually located far from the ocean.
- P. These volcanoes are centrally located around one city.
- R. The source of these volcanoes is along the West Coast.
- S. Volcanoes become inactive if they are under ocean water.

Farmers often use fertilizer on their farms. Fertilizer is also used in gardens. Some people argue that fertilizer is bad for the environment.

Which effect of using fertilizer **best** supports the argument that fertilizer is bad for the environment?

- M. Fertilizer stops soil erosion in fields.
- P. Fertilizer puts nutrients back into the soil.
- R. Fertilizer makes plants grow bigger and faster.
- S. Fertilizer disrupts the healthy balance of the soil.

The diagram shows a food chain.

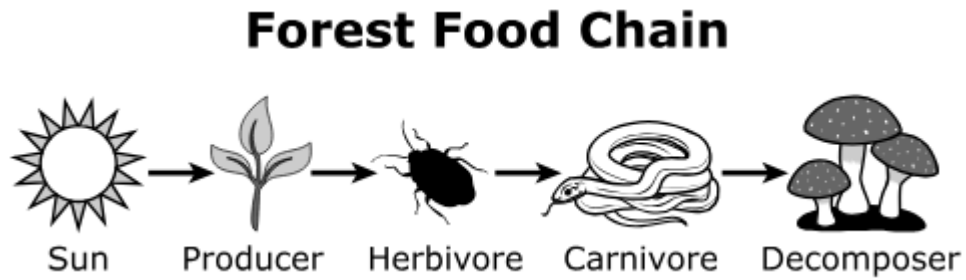
Food Chain

sunlight → small water plants → small water animals → fish → seals

Which **three** statements **best** describe how energy moves through this food chain?

- M. Seals get energy from fish.
- P. Fish get energy from seals.
- R. Fish get energy from small water animals.
- S. Small water plants get energy from sunlight.
- T. Small water plants get energy from small water animals

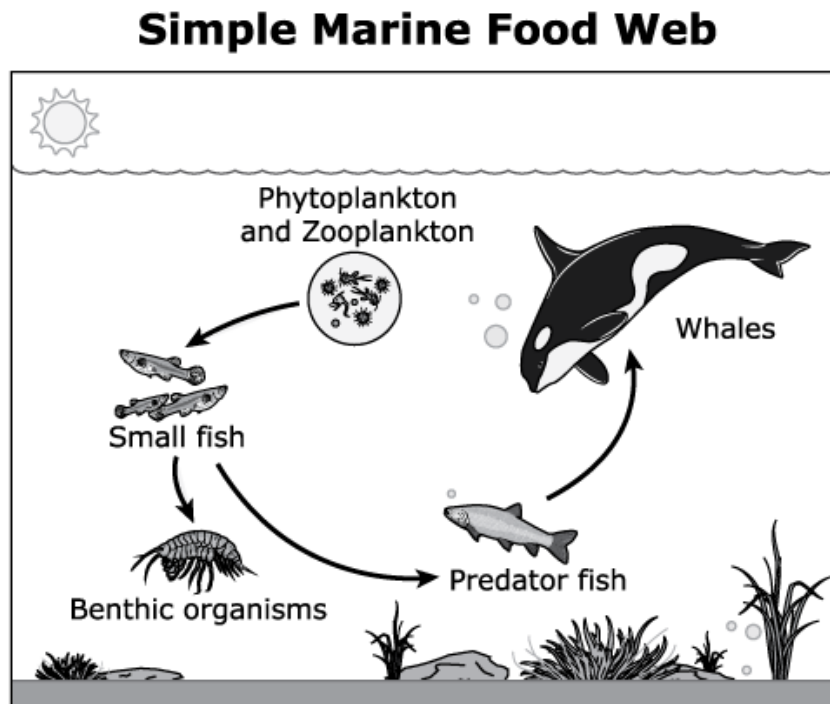
A forest food chain is shown.



Which **two** statements **best** describe the movement of energy in this forest food chain?

- A. The decomposer provides energy to the producer.
- B. The carnivore provides energy to the producer.
- C. The carnivore provides energy to the herbivore.
- D. The herbivore provides energy to the carnivore.
- E. The producer provides energy to the herbivore.

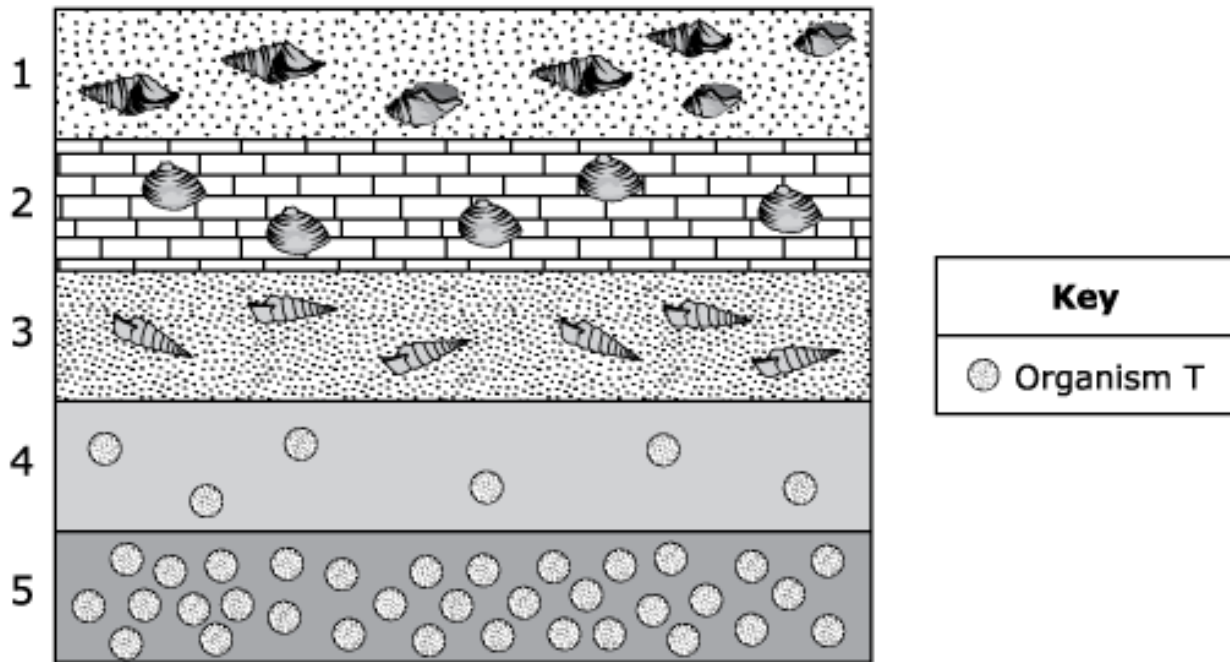
A simple marine food web is shown.



The small fish are important in this ecosystem because they

- M. give energy directly to plankton and benthic organisms.
- P. get energy from predator fish and benthic organisms.
- R. give energy to zooplankton and get energy from predator fish.
- S. get energy from both types of plankton and give energy to predator fish.

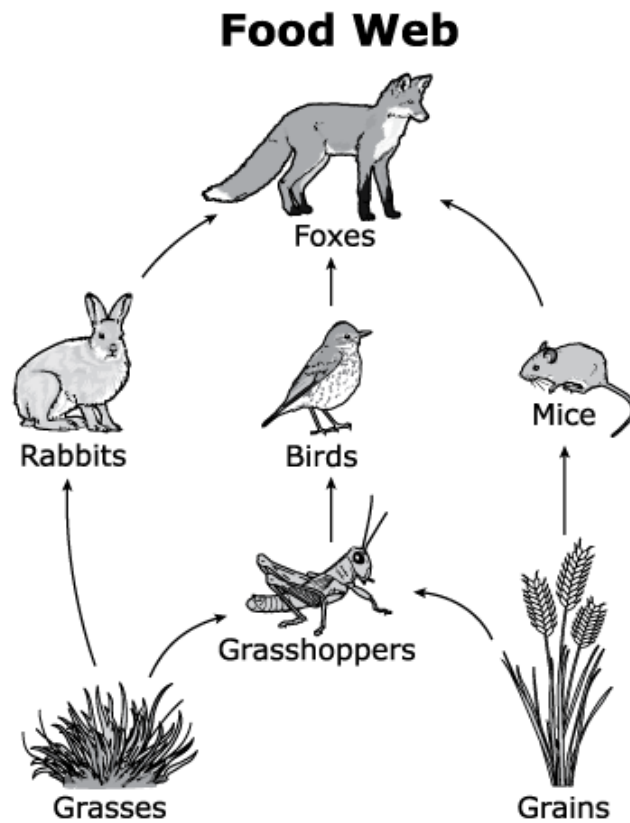
Organism T became extinct about 250 million years ago. The rock strata show fossil records for Organism T and some other organisms.



Because Organism T was found in rock layers 4 and 5, what conclusion can be made?

- M. Layer 1 is the oldest of the rock strata in the diagram.
- P. Layer 3 was formed less than 250 million years ago.
- R. The formation of layer 5 occurred less than 250 million years ago.
- S. Organism T existed on Earth for only a few years.

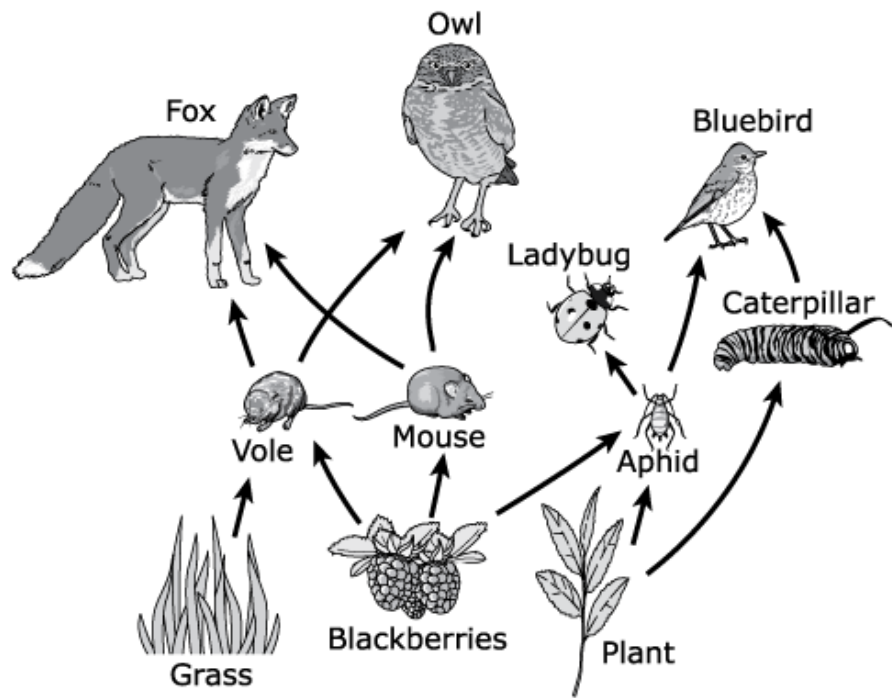
The diagram shows a food web. A food web shows feeding relationships among different organisms that live in the same habitat.



Suddenly, there were fewer grasshoppers. Based on this food web, which population of organisms will also decrease because of this change?

- M. Grasses
- P. Birds
- R. Mice
- S. Rabbits

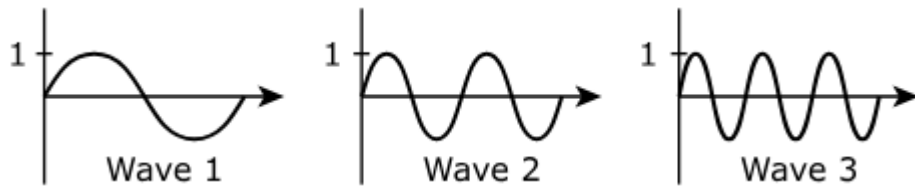
European starling birds are a species that was brought to the United States in the 1800s. They were used to help control insect populations. The starlings eat berries and insects such as caterpillars. A food web is shown.



Based on the food web, which **three** ways would the introduction of the starling **most likely** affect this food web?

- A. The caterpillar population would decrease.
- B. The mouse population would decrease.
- C. The bluebird population would decrease.
- D. The ladybug population would increase.
- E. The fox population would increase.

Students compared three models of waves. The models are shown.



Which statement correctly compares two of the models?

- M. Wave 1 has a smaller amplitude than Wave 2.
- P. Wave 2 has a bigger amplitude than Wave 3.
- R. Wave 2 has a longer wavelength than Wave 1.
- S. Wave 3 has a shorter wavelength than Wave 2.

A student looks at a spoon in a glass of water. The student observes that the spoon appears to be broken.



Why does the spoon look like it is broken?

- A. Light waves are bent by the water.
- B. Light waves are enlarged by the water.
- C. Light waves are absorbed by the water.
- D. Light waves are reflected by the water.

Metadata- Science Items

Page Number	Item Type	Key	TN Standards
1	MC	R	4.ESS2.2
2	MC	S	4.ESS3.2
3	MS	M, R, S	4.LS2.1
4	MS	D,E	4.LS2.1
5	MC	S	4.LS2.2
6	MC	P	4.ESS1.2
7	MC	P	4.LS2.3
8	MS	A,B,C	4.LS2.3
9	MC	S	4.PS4.1
10	MC	A	4.PS4.2
Metadata Definitions			
Item Type	MC = Multiple Choice MS = Multiple Select		
Key	Correct answer		
TN Standards	Primary educational standard assessed.		